Scientific Validation and Data Quality Assessment

SINTEF Ocean Real Vessel Data Challenge | TDT4259 Applied Data Science

25 October 2026 | Preliminary technical findings

1. Scope of validation

This assessment examines whether the available sensor records, feature definitions and descriptive analyses provide a defensible basis for the proposed **shaft mechanical power regression** study. It covers raw-data structure, identity and parsing checks, temporal alignment, feature suitability, unusual target measurements and the interpretation of descriptive correlations. **Model training and out-of-time predictive evaluation have not yet been performed.**

2. Checks and established findings

Area	Evidence and interpretation
Data inventory	**42,192,373 records** across thirteen sensor files; **41,079,968** belong to 2024 and **1,112,405** to January 2025.
Channel identification	37 registered QIDs; 36 observed in 2024 and 26 in January 2025. The shaft-power target is not available in January 2025.
Record integrity	No malformed short rows, numeric parsing failures or duplicate original `(timestamp, QID)` keys were identified in the full-file scan.
Parsed data types	All observed sensor values were parsed as finite binary64 numbers; the unobserved registered channel has no inferred measured datatype. Timestamps and QIDs are recorded as text.
Candidate input roles	**24 candidate channels** after excluding the target, six torquemeter-linked measurements, four constant-zero draft channels and sparse October water depth.
Draft channel behavior	**8,226,522 draft readings** across thirteen files are exactly zero; they are unsuitable as informative draft measurements.
October target availability	**88,304 raw target readings**, representing **22,088 of 44,640** completed one-minute intervals (49.48%).
Environmental overlap	All fourteen provider channels overlap **22,088 target minutes**, but only **369 distinct source timestamps** per channel contribute to those target-aligned minutes.
Source-specific overlap	Scavenging-air pressure has **22,087** October target-paired minutes and CAMS engine RPM **22,086**. Water depth has no target overlap in that month.
Noon Report dates	**366 distinct valid dates**, covering the 2024 calendar year.

The computational quality checklist records **33 satisfied assertions, zero failed assertions and three explicitly unresolved items**. These results confirm only the specified integrity and cross-check conditions. They do not establish that the underlying sensor values are free from calibration problems or that all aligned features were available operationally at their assigned timestamps.

3. Target measurement reliability

The strongest concern is the pattern of negative shaft-power values. **July** contains **28,544 negative readings** over fourteen dates, including three dates with exclusively negative readings and a near-contiguous sequence of **17,399**. **November** contains **656 negative readings** over eight dates, with values as low as approximately **−24,757 kW**. Negative power observed alongside positive CAMS engine RPM and water-relative speed provides additional reason to investigate the measurements, but does not independently establish the physical or instrument cause.

Consequently, July and November are reserved for **quality investigation rather than primary model fitting or headline evaluation**. The challenge documentation reports torquemeter miscalibration in part of the dataset; the exact failure periods have not been confirmed. **December also contains 166 negative power readings**, so any quality-filtered test

metric requires a policy defined before examining test prediction errors. Raw values should not be clipped or removed without a traceable rule and reported sensitivity analysis.

4. Temporal and statistical validity

Completed-minute alignment. Onboard readings are summarized over an interval ending at the labelled time; no later onboard measurement is required to describe that completed interval. Provider measurements are carried from the latest raw timestamp no more than sixty minutes earlier. This is a descriptive association rule, **not proof of real-time availability**, because provider valid time and issuance time may differ. The true timezone of the January 2025 timestamps is also unconfirmed.

Independence of evidence. Repeated target minutes and hourly provider measurements are serially dependent. Thus, target-paired minute counts must not be interpreted as independent statistical samples. The October correlation between scavenging-air pressure and power is about **+0.978** on **22,087 paired minutes**; it is an association, not evidence of causal contribution.

Speed versus power. October binned medians decrease across part of the observed speed range, despite the monotonic form of an illustrative cubic power law. The analysis does not determine whether loading, environmental conditions, operational regimes, the sensor measurements or a combination of these explains the decline. The cubic relationship is therefore **a physical reference, not a validated vessel-specific empirical law**.

Leakage control. Torquemeter-linked channels and close proxies are excluded from the candidate predictors. Other engine features have distinct registered source labels, but their measurement latency and availability remain to be established before operational use can be claimed. Circular direction variables require angle-aware preprocessing.

5. Evaluation design and decisions still required

The proposed evaluation uses **January–June 2024 for training, August for validation and December for untouched testing**. July and November are examined separately for target reliability; September and October are retained for coverage-stress investigations. This separation avoids random mixing of adjacent minutes and reserves a later period for a final assessment. No model selection based on December test errors should occur.

Three issues remain explicitly unresolved:

1. **Torquemeter reliability:** the exact mechanism and time limits of the suspect power measurements.
2. **Environmental timing:** whether a provider timestamp refers to issue time, valid time or another convention, including timezone interpretation.
3. **Operational signal availability:** the latency and independent availability of engine measurements at the point when the proposed estimate would be produced.

Two related limitations also affect interpretation: the cause of the unusual October speed–power pattern is undetermined, and the publication time of daily Noon Report values is not established. These conditions should be addressed through metadata, sensitivity analyses or conservative feature restrictions before claiming deployable performance.

6. Validation conclusion

The sensor inventory, data-type definitions, feature-role accounting, sampling denominators and descriptive findings are internally coherent enough to support **preliminary problem formulation and data understanding**. Significant target-quality anomalies and timestamp uncertainties remain visible rather than being treated as resolved. The next scientific decision is to establish defensible label-quality criteria and an inference-time feature policy, then assess actual generalization with chronological baselines and trained models. No predictive accuracy or operational savings can be inferred from the present descriptive analysis.